Indian Institute of Engineering Science and Technology, Shibpur
BTech 4th Semester End-Semester Examination, May 2026

Introduction to Data Science (CS2205)

Full Marks: 50 **Time: 3 Hours**

Attempt Question number 1 and any four from the rest. **(10+4×10)**

Q. No.	Body of the question
1	Select the correct answers. [10]
1(a)	Which of the following measures is commonly used to select the best attribute in a decision tree? i) Mean Squared Error ii) Information Gain iii) Correlation Coefficient iv) Jaccard Index
1(b)	Which visualization method is commonly used to identify outliers? i) Pie chart ii) Bar chart iii) Box plot iv) Line graph
1(c)	The main goal of regression is to: i) Predict categorical labels ii) Group similar data points iii) Predict continuous values iv) Reduce dimensionality
1(d)	Which measure is commonly used to assign points to clusters in K-Means? i) Manhattan distance only ii) Euclidean distance iii) Cosine similarity only iv) Hamming distance
1(e)	What is the main purpose of pruning in decision trees? i) Increase tree depth ii) Reduce training time iii) Avoid overfitting iv) Increase number of nodes
1(f)	Which method divides a continuous variable into bins of equal size? i) Equal-frequency binning ii) Clustering-based binning iii) Entropy-based binning iv) Equal-width binning
1(g)	Which of the following is NOT a preprocessing step? i) Feature scaling ii) Data cleaning iii) Model evaluation iv) Data transformation
1(h)	In regression, overfitting occurs when: i) Model performs well on both training and test data ii) Model is too simple iii) Model learns noise in the training data iv) Model ignores training data
1(i)	A major issue caused by class imbalance is: i) Improved generalization ii) Bias toward majority class iii) Faster training always iv) Reduced dataset size
1(j)	Which technique is used to convert categorical data into numerical form? i) Discretization ii) Encoding iii) Smoothing iv) Pruning
2(a)	Discuss the concept of entropy-based discretization. Explain its working procedure and illustrate with an example. [5]
2(b)	Describe the K-Nearest Neighbour (KNN) algorithm with example. Discuss the merits and demerits of the algorithm. [5]

3(a) State and prove Bayes' Theorem. Apply the theorem, clearly mentioning any assumptions or constraints, and write the Naïve Bayes Classification algorithm. [3+4]

3(b) What is the zero-probability problem in the Naïve Bayes Classification algorithm? Explain how it may lead to overfitting of the model and how this problem can be resolved. [3]

4(a) What is the class imbalance problem? What challenges does it present? [4]

4(b) Discuss the following methods used to handle the class imbalance problem:

(i) Tomek Links
(ii) SMOTE
(iii) SMOTE-ENN [6]

5(a) What is the 0.632 Bootstrap method? Discuss in detail how it is used for evaluating classification models. [5]

5(b) Suppose we have a test set consisting of 150 samples from a dataset with a given class distribution, as follows:

Class	Number of Samples
C1	60
C2	40
C3	50

The confusion matrix obtained after training and evaluating a classifier on this test set is provided, as follows:

		Actual Class		
		C1	C2	C3
Predicted class	C1	50	2	7
	C2	6	35	3
	C3	4	3	40

Compute the following:

(i) Accuracy of the classifier
(ii) Class-wise Precision, Recall, and F1-Score
(iii) Overall macro-averaged Precision, Recall, and F1-Score of the classifier

[5]

6 Write a short note on the following: [5×2]

(a) Principle Component Analysis (PCA)
(b) K-Means Clustering Algorithm